

Statistical Mechanics

Mid-Term Examination, 2026

Harish-Chandra Research Institute, Prayagraj

- Q1. Imagine throwing a die and attempting to determine the probability distribution of the outcomes. There are six possible outcomes,

$$\alpha = 1, 2, 3, 4, 5, 6,$$

with probabilities p_α .

- (a) If we know absolutely nothing about the die and adopt the principle of maximum entropy as a guiding principle, what should be our *a priori* expectation for the probabilities p_α ? What do we then expect the average outcome

$$\langle \alpha \rangle = \sum_{\alpha=1}^6 \alpha p_\alpha$$

to be? [3]

- (b) Suppose someone has performed very many throws and informs us that the average outcome is $\langle \alpha \rangle = 3.667$. Use the principle of maximum entropy to determine the probabilities p_α for $\alpha = 1, \dots, 6$. [7]

$$3.667 = 1 \langle p_\alpha \rangle + 2 \langle p_\alpha \rangle + \dots$$

- Q2. Density of states.

- (a) Consider a particle living in a 2D box. What is the density of states $g(k)$ for it? [2]

- (b) What is $g(k)$ for a particle in a 1D box? [2]

- (c) Calculate the density of states and Fermi energy in a d -dimensional box. [4]

$$k^2, k^2$$

Hint: In d spatial dimensions, the surface area of the unit $(d-1)$ -sphere is given by

$$S_{d-1} = \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})}.$$

- Q3. Consider an ideal gas of bosons whose density of states is given by $g(E) = CE^{(\alpha-1)}$ for some constants C and $\alpha > 1$. Derive an expression for the critical temperature T_c , below which the gas experiences Bose-Einstein condensation. Elaborate the concept of critical temperature with physical explanation. [10]

$$\mu=0$$

- Q4. By shining an intense laser beam on a semiconductor, one can create a metastable collection of electrons (charge $-e$, effective mass m_e) and holes (charge $+e$, effective mass m_h) in the bulk. The oppositely charged particles may pair up (as in a hydrogen atom) to form a gas of excitons, or they may dissociate into an electron-hole plasma. We examine a simplified model of this process.

$$\mu = -T \frac{\partial S}{\partial N}$$
$$S \propto k \frac{\partial}{\partial T} \left(\ln(Z) \right)$$

$$\epsilon' \sqrt{\kappa c}$$

- (a) Calculate the free energy of a gas composed of N_e electrons and N_h holes at temperature T , treating them as classical non-interacting particles with masses m_e and m_h . [5]
- (b) When an electron and a hole bind to form an exciton, the energy is lowered by an amount ϵ . (The binding energy of a hydrogen-like exciton is approximately

$$\epsilon \approx \frac{m_r e^4}{2\hbar^2 \epsilon^2},$$

where ϵ is the dielectric constant and the reduced mass m_r is $m_r^{-1} = m_e^{-1} + m_h^{-1}$. Calculate the free energy of a gas of N_p excitons, treating them as classical non-interacting particles with mass [5]

$$m = m_e + m_h.$$

- (c) Calculate the chemical potentials μ_e , μ_h , and μ_p of the electron, hole, and exciton gases, respectively. [3]
- (d) Using the condition for chemical equilibrium,

$$\mu_e + \mu_h = \mu_p,$$

derive the relation between the densities of electrons, holes, and excitons. [2]

- (e) At high temperature T , find the density of excitons n_p as a function of the total excitation density [5]

$$n \approx n_e + n_h.$$

- Q5. Consider an isolated composite system consisting of two weakly interacting subsystems A and B that can exchange energy but not particles or volume. The total energy is fixed,

$$E = E_A + E_B.$$

Let $\Omega_A(E_A)$ and $\Omega_B(E_B)$ denote the numbers of microstates available to the subsystems at energies E_A and E_B , respectively. For a given division of energy E_A , the total number of accessible microstates is

$$\Omega(E_A) = \Omega_A(E_A) \Omega_B(E - E_A).$$

Since the composite system is isolated, all microstates at fixed E are equally probable. However, different values of E_A correspond to different multiplicities $\Omega(E_A)$, and the equilibrium value E_A^* is the one that maximizes $\Omega(E_A)$.

- (a) Show that the equilibrium condition is [7]

$$\left(\frac{\partial S_A}{\partial E_A} \right) = \left(\frac{\partial S_B}{\partial E_B} \right), \quad \frac{\partial}{\partial}$$

where

$$S_i(E_i) = k_B \ln \Omega_i(E_i).$$

- (b) Interpret this condition physically. [3]

Statistical Mechanics

Re-examination (Mid-Term), 2026

Harish-Chandra Research Institute, Prayagraj

- Q1.** Consider a classical gas of non-interacting particles moving in one dimension. At time $t = 0$, the particles are confined to the spatial region $0 \leq x \leq L$, but their momenta can take any value. The initial single-particle distribution is

$$f(x, p, 0) = A \exp\left(-\frac{p^2}{2mk_B T}\right) \quad \text{for } 0 \leq x \leq L,$$

and

$$f(x, p, 0) = 0 \quad \text{otherwise,}$$

where A is a normalization constant. At time $t = 0$, the walls are suddenly removed and the particles evolve freely with Hamiltonian

$$H = \frac{p^2}{2m}.$$

- (a) Write down Hamilton's equations for the motion and determine $x(t)$ and $p(t)$ in terms of the initial values x_0 and p_0 .
- (b) Using Liouville's theorem, show that the phase-space distribution remains constant along trajectories. Hence obtain $f(x, p, t)$ at later times.
- (c) Find the particle density

$$n(x, t) = \int dp f(x, p, t),$$

$$\frac{d\hat{e}}{dt} = \{\hat{e}, H\}$$

and describe qualitatively how the gas expands in real space.

- (d) Does the phase-space density decrease during the expansion? Explain carefully. How is this consistent with the intuitive idea that the gas is "spreading out"?
- (e) Compute the Gibbs entropy

$$S = -k_B \int dx dp f(x, p, t) \ln f(x, p, t),$$

and show that it remains constant in time under Hamiltonian evolution.

- (f) Explain why this result does not contradict the thermodynamic expectation that entropy increases during free expansion. What additional physical idea is needed?

- Q2.** Dirac fermions are non-interacting particles of spin $1/2$. The one-particle states come in pairs of positive- and negative-energy branches,

$$E_{\pm}(\vec{k}) = \pm \sqrt{m^2 c^4 + \hbar^2 k^2 c^2},$$

independent of spin.

- (a) For any fermionic system of chemical potential μ , show that the probability of finding an occupied state of energy $\mu + \delta$ is the same as that of finding an unoccupied state of energy $\mu - \delta$. (δ is any constant energy.)
- (b) At zero temperature all negative-energy Dirac states are occupied and all positive-energy ones are empty, i.e. $\mu(T = 0) = 0$. Using the result in (a), find the chemical potential at finite temperatures T .
- (c) Show that the mean excitation energy of this system at finite temperature satisfies

$$E(T) - E(0) = 4V \int \frac{d^3 \vec{k}}{(2\pi)^3} \frac{\mathcal{E}_+(\vec{k})}{\exp(\beta \mathcal{E}_+(\vec{k})) + 1}.$$

- (d) Evaluate the integral in part (c) for massless Dirac particles (i.e. for $m = 0$).
- (e) Calculate the heat capacity, C_V , of such massless Dirac particles.
- (f) Describe the qualitative dependence of the heat capacity at low temperature if the particles are massive.

Q3. *Bose gas in d dimensions:* Consider a gas of non-interacting (spinless) bosons with an energy spectrum $\epsilon = p^2/2m$, contained in a box of “volume” $V = L^d$ in d dimensions.

- (a) Calculate the grand potential $\mathcal{G} = -k_B T \ln \mathcal{Q}$, and the density $n = N/V$, at a chemical potential μ . Express your answers in terms of d and $f_m^+(z)$, where $z = e^{\beta\mu}$, and

$$f_m^+(z) = \frac{1}{\Gamma(m)} \int_0^\infty \frac{x^{m-1}}{z^{-1}e^x - 1} dx.$$

(Hint: Use integration by parts on the expression for $\ln \mathcal{Q}$.)

- (b) Calculate the ratio PV/E , and compare it to the classical value.
- (c) Find the critical temperature, $T_c(n)$, for Bose-Einstein condensation.
- (d) Calculate the heat capacity $C(T)$ for $T < T_c(n)$.
- (e) Find the ratio, $C_{\max}/C(T \rightarrow \infty)$, of the maximum heat capacity to its classical limit, and evaluate it in $d = 3$.
- (f) How does the above calculated ratio behave as $d \rightarrow 2$? In what dimensions are your results valid? Explain.

Q4. Consider a single quantum harmonic oscillator of frequency ω in contact with a heat bath at temperature T . Its energy levels are

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

The system is described by the canonical ensemble.

- (a) Write down the partition function and show that it can be written as

$$Z = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}}.$$

$$\int u dv = uv - \int v du$$

$$\int x^2 dx = \frac{x^3}{3}$$

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- (b) Show that the mean energy is

$$\langle E \rangle = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}.$$

- (c) Obtain an explicit expression for the energy fluctuation $(\Delta E)^2$.
(d) Show directly that

$$(\Delta E)^2 = k_B T^2 C,$$

where $C = \frac{\partial \langle E \rangle}{\partial T}$ is the heat capacity of the oscillator.

- (e) Study the two limits: low-temperature limit $k_B T \ll \hbar\omega$ and high-temperature limit $k_B T \gg \hbar\omega$. Show that in the high-temperature limit your results reduce to the classical oscillator result.
(f) Evaluate the relative fluctuation $\frac{\Delta E}{\langle E \rangle}$ in the low- and high-temperature limits, and comment briefly on why fluctuations remain high.

Statistical Mechanics Assignment – I

Q1. Consider a system containing four particles whose potential energy is

$$V = \frac{a}{2} (9x_1^2 + 8x_1x_2 + 3x_2^2 + x_3^2) + \frac{b}{3}x_3^3 + \frac{b_4}{4}x_3^4; \quad a \geq 0; b \geq 0; c \geq 0$$

Calculate average energy, specific heat and average value of x_3 of this system at a temperature T when $b = c = 0$. Now for small values of b and c calculate average of x_3 as temperature changes considering leading order in b and c .

Q2. Consider a physical system that can be found in only two possible configurations, labeled 1 and 2. No prior knowledge about their likelihoods is available. To assign probabilities empirically, you perform a large number $N \gg 1$ of identical independent trials of a binary experiment (for example, repeated measurements or coin flips).

Let N_1 and N_2 denote the number of times outcomes 1 and 2 occur, respectively, with

$$N_1 + N_2 = N, \quad p_1 = \frac{N_1}{N}, \quad p_2 = \frac{N_2}{N}.$$

(I) Most likely empirical probability assignment

- (a) Determine the multiplicity $W(N_1, N_2)$, i.e., the number of distinct sequences of outcomes that lead to the same counts $\{N_1, N_2\}$.
- (b) Find the value of N_1 (or equivalently p_1) for which this multiplicity is maximal, and show that the most probable empirical assignment corresponds to equal probabilities for the two states.
- (c) Compute the Gibbs (Shannon) entropy associated with this most probable distribution.

(II) Sharpness of the multiplicity peak for large samples

Let the number of occurrences of outcome 1 deviate slightly from its most probable value by writing

$$N_1 = \frac{N}{2} - m, \quad N \gg m \gg 1.$$

- (a) Using Stirling's approximation, show that the multiplicity near the maximum satisfies

$$\frac{W(m)}{W(0)} \approx \exp\left(-\frac{2m^2}{N}\right),$$

where $W(0)$ denotes the multiplicity at the most probable outcome.

- (b) Deduce that the relative fluctuation of the inferred probability,

$$\delta p = \frac{m}{N},$$

scales as $\delta p \sim N^{-1/2}$, demonstrating that the peak becomes increasingly sharp as the sample size grows.

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- Q3.** Consider a low density gas, where the interaction between two gas molecule is $u(r)$ (given below). Furthermore assume the equation of state of this gas can be written as

$$\frac{p}{k_B T} = \frac{N}{V} + B \left(\frac{N}{V} \right)^2 + C \left(\frac{N}{V} \right)^3.$$

Here p denotes pressure, N is number of particle and V denotes volume. k_B is Boltzman constant. Following the calculations shown in tutorial 1, calculate B and C . Take $u(r)$ as

$$u(r) = \begin{cases} \infty, & 0 \leq r < r_0 \\ -u_0 \left(\frac{r_0}{r} \right)^6, & r \geq r_0 \end{cases}$$

- Q4.** Consider a gas of non-interacting (spinless) bosons with single-particle energy

$$\varepsilon = c |\vec{p}|,$$

contained in a box of “volume” $V = L^d$ in d spatial dimensions.

- (a) Calculate the grand potential

$$\mathcal{G} = -k_B T \ln \mathcal{Q},$$

and the particle density

$$n = \frac{N}{V},$$

at chemical potential μ .

Express your answers in terms of the spatial dimension d and the function

$$f_+^{(m)}(z) = \frac{1}{\Gamma(m)} \int_0^\infty \frac{x^{m-1}}{z^{-1}e^x - 1} dx, \quad z = e^{\beta\mu}.$$

- (b) Calculate the gas pressure P and the total energy E , and compare the ratio

$$\frac{E}{PV}$$

with the classical ideal-gas value.

- (c) Determine the critical temperature $T_c(n)$ for Bose–Einstein condensation and identify the spatial dimensions in which a condensation transition occurs.
- (d) What is the temperature dependence of the heat capacity $C(T)$ for $T < T_c(n)$?

- Q5.** A quantum harmonic oscillator of angular frequency ω has energy levels

$$E_n = \omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

(I) Single oscillator thermodynamics

- (a) Determine the canonical partition function $Z(T)$ of the oscillator at temperature T .

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- (b) Using Z , obtain expressions for the mean energy $E(T)$ and the entropy $S(T)$.

(II) Einstein model of a crystalline solid

Einstein proposed a simplified model of a solid in which each atom oscillates independently about its equilibrium position with the same characteristic frequency ω . A crystal of N atoms therefore behaves as a collection of $3N$ identical quantum harmonic oscillators.

- (a) Using the result for a single oscillator, compute the total heat capacity $C_V(T)$ of the solid.
- (b) Show that in the high-temperature limit $k_B T \gg \omega$, the model reproduces the classical Dulong–Petit law

$$C_V = 3Nk_B.$$

(III) Low-temperature behavior

Experiments show that at very low temperatures the heat capacity of many solids behaves as

$$C_V \propto T^3.$$

Examine the low-temperature prediction of the Einstein model and discuss whether it agrees with the observed behavior. If not, briefly comment on the physical reason for the discrepancy.

SM: Assignment

Topic: Phase Transition

Deadline: May 6, 2026

Q1. Consider the van der Waals equation of state

$$p = \frac{k_B T}{v - b} - \frac{a}{v^2}, \quad (1)$$

where $v = V/N$ is the volume per particle, and $a, b > 0$ are constants. Using this equation, discuss the liquid–gas transition and answer the following:

- (i) Sketch qualitatively the p - v isotherms for the three cases $T > T_c$, $T = T_c$, and $T < T_c$. Explain why the low-temperature isotherm develops a non-monotonic region. Identify the gas-like branch, liquid-like branch, and unstable branch. Explain why the condition

$$\left(\frac{\partial p}{\partial v} \right)_T > 0 \quad (2)$$

signals instability. [5]

- (ii) Show that the critical point is determined by

$$\left(\frac{\partial p}{\partial v} \right)_T = 0, \quad \left(\frac{\partial^2 p}{\partial v^2} \right)_T = 0. \quad (3)$$

Hence derive expressions for T_c , v_c , and p_c . [6]

- (iii) For $T < T_c$, explain why the system does not actually follow the full oscillating van der Waals isotherm. State the conditions for coexistence of liquid and gas. Starting from chemical equilibrium, derive the Maxwell equal-area construction and explain its physical meaning. [7]

- (iv) Explain the distinction between the unstable region, the metastable region, and the coexistence region. Define the spinodal curve and the coexistence curve. Briefly explain the meanings of supercooled vapour and superheated liquid. [4]

- (v) Using equality of Gibbs free energy per particle in the two phases, derive the Clausius–Clapeyron equation

$$\frac{dp}{dT} = \frac{L}{T(v_{\text{gas}} - v_{\text{liquid}})}, \quad (4)$$

where L is the latent heat per particle. Explain why the liquid–gas transition below T_c is a first-order phase transition. [6]

- (vi) Assume that L is constant, $v_{\text{gas}} \gg v_{\text{liquid}}$, and the gas obeys the ideal gas law. Under these assumptions, solve the Clausius–Clapeyron equation approximately and show that

$$p = p_0 e^{-L/(k_B T)}. \quad (5)$$

State clearly the approximations used and their physical significance. [3]

(vii) Define the reduced variables

$$\bar{T} = \frac{T}{T_c}, \quad \bar{v} = \frac{v}{v_c}, \quad \bar{p} = \frac{p}{p_c}. \quad (6)$$

Rewrite the van der Waals equation in terms of reduced variables. Hence explain the law of corresponding states. Also derive the compressibility ratio

$$\frac{p_c v_c}{k_B T_c} = \frac{3}{8}. \quad (7)$$

Comment briefly on why this is only approximately valid for real gases. [4]

(viii) Using the van der Waals description, show that near the critical point:

$$v_{\text{gas}} - v_{\text{liquid}} \sim (T_c - T)^{1/2}, \quad (8)$$

$$p - p_c \sim (v - v_c)^3, \quad (9)$$

$$\kappa \sim (T - T_c)^{-1}. \quad (10)$$

Explain what is meant by critical exponents, and briefly comment on why van der Waals theory does not give the experimentally correct values. [5]

Instructions: Consult with any standard text book. I suggest to arrange for a tutorial on this topic, as well.

Q2. Consider a magnetic system described by the Landau free-energy density

$$f(m) = a(T - T_0)m^2 - bm^4 + cm^6 - Bm, \quad a, b, c > 0, \quad (11)$$

where m is the magnetization order parameter, T is the temperature, and B is the external magnetic field. This model is known to describe a first-order magnetic phase transition for suitable parameter regimes. Using this free energy, study the equilibrium, metastable, and hysteretic behaviour of the system.

(i) For $B = 0$, determine the stationary points of the free energy by solving

$$\frac{df}{dm} = 0.$$

Show that the system can have a minimum at $m = 0$ and two additional symmetry-related minima at nonzero m . Explain under what condition these nonzero solutions exist. [5]

(ii) Show that the first-order transition temperature T_t is obtained by equating the free energies of the ordered and disordered minima. Derive an expression for T_t in terms of a, b, c , and determine the corresponding jump in magnetization at the transition. [7]

(iii) Show that the second derivative

$$\frac{d^2 f}{dm^2}$$

determines the local stability of a stationary point. Using this, identify which solutions are stable, metastable, or unstable. Explain clearly the distinction between these three cases in terms of the free-energy landscape. [5]

- (iv) The limits of metastability are determined by the spinodal conditions

$$\frac{df}{dm} = 0, \quad \frac{d^2f}{dm^2} = 0.$$

For $B = 0$, derive the spinodal temperatures associated with the disappearance of the ordered and disordered metastable states. Sketch the qualitative temperature dependence of the free-energy profile in the different regimes. [7]

- (v) Now consider $T < T_t$ and switch on a magnetic field $B \neq 0$. Explain qualitatively how the term $-Bm$ tilts the free-energy landscape. Show that one of the two ordered minima is favoured depending on the sign of B . [4]

Q3. Consider a 2D ferromagnetic Ising model on an $L \times L$ square lattice with Hamiltonian

$$H = -J \sum_{\langle ij \rangle} s_i s_j - \sum_i B(x_i) s_i, \quad s_i = \pm 1, \quad (12)$$

where $J > 0$, and the external magnetic field varies linearly in space,

$$B(x) = gx. \quad (13)$$

Here g is a constant field gradient, and x denotes the horizontal coordinate of a lattice site. Use mean-field theory to study how the usual ferromagnetic transition is modified by the field gradient.

- (i) Briefly recall the mean-field approximation for the Ising model. Using it, show that the local magnetization $m(x) \equiv \langle s_i \rangle$ satisfies the self-consistency equation

$$m(x) = \tanh[\beta(Jqm(x) + gx)], \quad (14)$$

where $q = 4$ for the square lattice. Explain why the magnetization must now depend on position. [5]

- (ii) Show that in the absence of gradient ($g = 0$), the above equation reduces to the standard mean-field equation for the uniform Ising model. Hence recover the mean-field critical temperature

$$k_B T_c = Jq. \quad (15)$$

[4]

- (iii) For $T > T_c$, assume that the magnetization is small and linearize the self-consistency equation. Show that

$$m(x) \approx \frac{\beta gx}{1 - \beta Jq}. \quad (16)$$

Explain why the local magnetic response becomes very large as $T \rightarrow T_c^+$. [5]

- (iv) Using your result above, sketch qualitatively the magnetization profile $m(x)$ across the sample for $T > T_c$. Explain why the field gradient removes the possibility of a single sharp bulk transition for the entire sample. [4]

(v) Now consider $T < T_c$. At each position x , the local mean-field equation is still

$$m(x) = \tanh[\beta(Jqm(x) + gx)]. \quad (17)$$

Explain qualitatively why:

- sites with $x > 0$ favour positive magnetization,
- sites with $x < 0$ favour negative magnetization,
- the central region near $x = 0$ is the most sensitive to the transition.

Describe the expected shape of $m(x)$ below T_c . [5 marks]

(vi) For $T < T_c$ and small $|gx|$, expand the mean-field equation to cubic order in $m(x)$:

$$m \approx \beta(Jqm + gx) - \frac{1}{3}\beta^3(Jqm + gx)^3. \quad (18)$$

Use this to show that near $x = 0$, the magnetization satisfies a local cubic equation of Landau type. Explain how this reproduces the usual mean-field spontaneous magnetization when $g \rightarrow 0$. [7]

(vii) Discuss the limit $g \rightarrow 0$. Show that one recovers the standard uniform ferromagnetic transition, with a spatially uniform order parameter. Explain why any nonzero g rounds the transition spatially. [4]

Statistical Mechanics

End-Term, 2026

Harish-Chandra Research Institute, Prayagraj

Instruction: Provide detailed derivations.

1. Consider the velocity of a gas particle in one dimension,

$$-\infty < v < \infty.$$

- (a) Find the unbiased probability density $p_1(v)$, subject only to the constraint that the average speed is c , that is,

$$\langle |v| \rangle = c.$$

[5 marks]

- (b) Now find the probability density $p_2(v)$, given only the constraint of average kinetic energy,

$$\left\langle \frac{mv^2}{2} \right\rangle = \frac{mc^2}{2}.$$

[5 marks]

2. Consider a single-particle classical ensemble in one dimension. At $t=0$, each realization is prepared at $q=0$, the center of a hard-wall box. The initial phase-space density is

$$\rho(q, p, t = 0) = \delta(q)f(p),$$

where

$$f(p) = \frac{1}{\sqrt{2\pi mk_B T}} \exp\left(-\frac{p^2}{2mk_B T}\right).$$

Each realization subsequently evolves freely.

- (a) Starting from Liouville's equation, derive $\rho(q, p, t)$ and sketch it in the (q, p) plane. [5 marks]

- (b) Derive the expressions for the averages $\langle q^2 \rangle$ and $\langle p^2 \rangle$ at $t > 0$. [5 marks]

3. Irreversible processes

- (a) Consider two substances, initially at temperatures T_1^0 and T_2^0 , coming to equilibrium at a final temperature T_f through heat exchange. By relating the direction of heat flow to the temperature difference, show that the change in the total entropy,

$$\Delta S = \Delta S_1 + \Delta S_2 \geq \int_{T_1^0}^{T_f} \frac{dQ_1}{T_1} + \int_{T_2^0}^{T_f} \frac{dQ_2}{T_2} = \int \frac{T_1 - T_2}{T_1 T_2} dQ,$$

must be positive. Hence argue that, in a closed system, equilibrium is characterized by the maximum value of the entropy S .

[5 marks]

- (b) Now consider a gas with adjustable volume V , and diathermal walls, embedded in a heat bath of constant temperature T and fixed pressure P . The change in entropy of the bath is given by

$$\Delta S_{\text{bath}} = \frac{\Delta Q_{\text{bath}}}{T} = -\frac{\Delta Q_{\text{gas}}}{T} = -\frac{1}{T} (\Delta E_{\text{gas}} + P\Delta V_{\text{gas}}).$$

By considering the change in entropy of the combined system, establish that the equilibrium of a gas at fixed T and P is characterized by the minimum of the Gibbs free energy

$$G = E + PV - TS.$$

[5 marks]

4. Consider two identical, non-interacting particles confined in a two-dimensional isotropic harmonic trap. The single-particle energy levels are given by

$$\varepsilon_{n_x, n_y} = \hbar\omega(n_x + n_y + 1), \quad n_x, n_y = 0, 1, 2, \dots$$

In this problem, consider only the **first three** single-particle energy levels.

- (a) First consider two identical bosons in the trap. Enumerate all allowed two-particle microstates using the above three single-particle levels. For each microstate, write down its total energy.

[5 marks]

- (b) Using the microstate counting in part (a), construct the canonical partition function $Z_B(\beta)$ and then obtain the Helmholtz free energy F_B , the mean energy U_B , and the heat capacity C_B .

[5 marks]

- (c) Now consider two identical spinless fermions in the same trap and perform the same set of calculations provided in parts (a) and (b) [Just replace bosons by fermions]. Here denote canonical partition function, Helmholtz free energy, the mean energy and heat capacity by $Z_F(\beta)$, F_F , U_F and C_F , respectively.

[10 marks]

5. Consider an ideal gas of electrons in three dimensions in a uniform magnetic field B . Ignore the orbital motion and keep only the spin coupling. The single-particle energy is

$$\epsilon_{\vec{k}, s} = \frac{\hbar^2 k^2}{2m} + \mu_B B s, \quad s = \pm 1,$$

where $\mu_B = |e|\hbar/2mc$. Let $z = e^{\beta\mu}$ be the fugacity.

- (a) Show that the spin magnetization is

$$M = -\mu_B (N_{\uparrow} - N_{\downarrow}) = -\frac{\mu_B V}{\lambda^3} [f_{3/2}(ze^{-\beta\mu_B B}) - f_{3/2}(ze^{+\beta\mu_B B})],$$

where

$$f_{3/2}(z) = \frac{2}{\sqrt{\pi}} \int_0^\infty dx \frac{x^{1/2}}{z^{-1}e^x + 1}$$

is the Fermi-Dirac integral of order $3/2$.

[5 marks]

(b) In the high-temperature limit $z \ll 1$, show that $M \simeq \mu_B N \tanh(\beta \mu_B B)$.

Deduce the zero-field susceptibility and show that it satisfies Curie's law,
 $\chi = \frac{N \mu_B^2}{k_B T}$. [5 marks]

(c) Now perform an analogous analysis [both part (a) and (b)] by considering a gas of non-interacting spin-1 bosons, where the spin quantum number s takes the three possible values

$$s = -1, 0, +1.$$

[10 marks]

6. Consider a lattice of spin-1 variables, $S_i = 0, \pm 1$ described by the ferromagnetic Hamiltonian

$$H = -\frac{J}{2N} \sum_{i,j} S_i S_j - B \sum_i S_i, \quad J > 0.$$

Here B is a uniform magnetic field.

(a) Using mean-field theory, show that the self-consistency equation for the magnetization is

$$m = \frac{2 \sinh[\beta(Jm + B)]}{1 + 2 \cosh[\beta(Jm + B)]}.$$

Explain why, for $B \neq 0$, any nonzero magnetization is field-induced rather than spontaneous. [5 marks]

(b) By expanding the self-consistency equation for small m , show that a nonzero solution can appear spontaneously even in the absence of the magnetic field below a critical temperature T_c . Derive expression for T_c . Hence explain why spontaneous magnetization requires interactions in this model. [5 marks]

(c) Near T_c , expand the mean-field equation in m . Show that for $T < T_c$,

$$m \sim (T_c - T)^{\beta_{\text{MF}}},$$

and determine the mean-field critical exponent β_{MF} . [5 marks]

(d) Now set $T = T_c$ and keep a small uniform field $B \neq 0$. Expand the self-consistency equation for small m and small B . Show that the critical isotherm has the form

$$m \sim B^{1/\delta_{\text{MF}}},$$

and determine the mean-field exponent δ_{MF} . [5 marks]